

Exp : 2 Write the Python Program to solve 8-Queen

Aim:

To write a python program to solve 8-Queen problem.

Algorithm:

Step1: Start with an empty board of size 8x8.

Step2: Place the first queen in the first row and the first column.

Step3: For each subsequent row, check all possible columns to place the queen, and backtrack if a valid position cannot be found.

Step4: A valid position is a column where no other queens are in the same row, column, or diagonal.

Step5: If no solution is found, return failure.

Program:

```
N = 8
```

```
board = [[0 for _ in range(N)] for _ in range(N)]
```

```
def is_safe(board, row, col):
```

```
    # Check left side
```

```
    for i in range(col):
```

```
        if board[row][i] == 1:
```

```
            return False
```

```
    # Check upper diagonal
```

```
    i, j = row, col
```

```
    while i >= 0 and j >= 0:
```

```
        if board[i][j] == 1:
```

```
            return False
```

```
        i -= 1
```

```
        j -= 1
```

```
# Check lower diagonal
```

```
i, j = row, col
```

```
while i < N and j >= 0:
```

```
    if board[i][j] == 1:
```

```
        return False
```

```
    i += 1
```

```
    j -= 1
```

```
return True
```

```
def solve(board, col):
```

```
    if col >= N:
```

```
        return True
```

```
    for row in range(N):
```

```
        if is_safe(board, row, col):
```

```
            board[row][col] = 1
```

```
            if solve(board, col + 1):
```

```
                return True
```

```
            board[row][col] = 0
```

```
    return False
```

```
if solve(board, 0):
```

```
    print("Solution:")
```

```
    for row in board:
```

```
        print(row)
else:
    print("No solution exists")
```

Output :

```
>>> =====
>>>
Solution:
[1, 0, 0, 0, 0, 0, 0, 0]
[0, 0, 0, 0, 0, 0, 1, 0]
[0, 0, 0, 0, 1, 0, 0, 0]
[0, 0, 0, 0, 0, 0, 0, 1]
[0, 1, 0, 0, 0, 0, 0, 0]
[0, 0, 0, 1, 0, 0, 0, 0]
[0, 0, 0, 0, 0, 1, 0, 0]
[0, 0, 1, 0, 0, 0, 0, 0]
>>>
```

Result :

Hence, the simple python program for 8- Queen problem is done